Genotype	Dominant/ recessive	Modification	Sex
ba/ba ts/ts	Double recessive	Lacks silk on the stalk, but transformed tassel to pistil	Rudimentary female
ba/ba ts^+/ts^+	Recessive and dominant	Lacks silk and have tassel	Male
ba^+/ba^+ ts^+/ts^+	Double dominant	Have both tassel and cob	Monoecious
ba^+/ba^+ ts/ts	Dominant and recessive	Bears cob and lacks tassel	Normal female

Table 3.6: Sex determination in Maize (Superscript (+) denotes dominant character)

3.5 Mutation



Mutant Leaf

Genetic variation among individuals provides the raw material for the ultimate source of evolutionary changes. Mutation and recombination are the two major processes responsible for genetic variation. A sudden change in the genetic material of an organisms is called **mutation**. The term mutation was introduced by **Hugo de Vries** (1901) while he has studying on the plant, evening primrose (***Oenothera lamarkiana***) and proposed **'Mutation theory'**. There are two broad types of changes in genetic material. They are point mutation and chromosomal mutations. Mutational events that take place within individual genes are called gene mutations or point mutation, whereas the changes occur in structure and number of chromosomes is called chromosomal mutation. Agents which are responsible for mutation are called **mutagens,** that increase the rate of mutation. Mutations can occur either spontaneously or induced. The production of mutants through exposure of mutagens is called mutagenesis, and the organism is said to be ***mutagenized.***

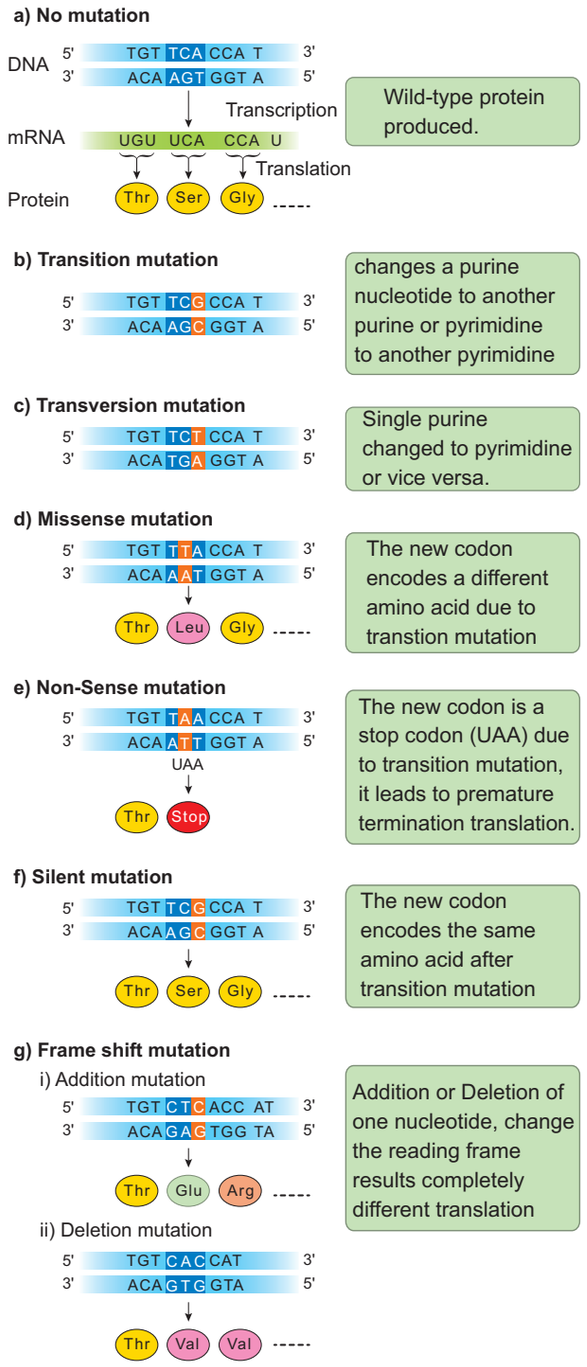


Figure: 3.9 Types of point mutation

3.5.1 Types of mutation

Let us see the two general classes of gene mutation:

TN_GOVT_BOTANY_XII_PAGES_052-081 CH 03.indd 60 28-12-2022 15:41:18